

8253/8254 Timer and Counters

- The 8253/54 solves one of most common problem in any microcomputer system, the **generation of accurate time delays** under software control.
- Instead of setting up timing loops in system software, the programmer **configures** the 8253/54 to match this requirements, **initializes one of the counters** of the 8253/54 with the desired quantity, then **upon command** the 8253/54 will count out the delay and **interrupt** the CPU when it has completed its tasks.

Some of the counter/timer functions common to microcomputers which can be implemented with the 8254 are:

- Real time clock
- Event-counter
- Digital one-shot
- Programmable rate generator
- Square wave generator

- Difference between 8253 and 8254
- The following table differentiates the features of 8253 and 8254 –

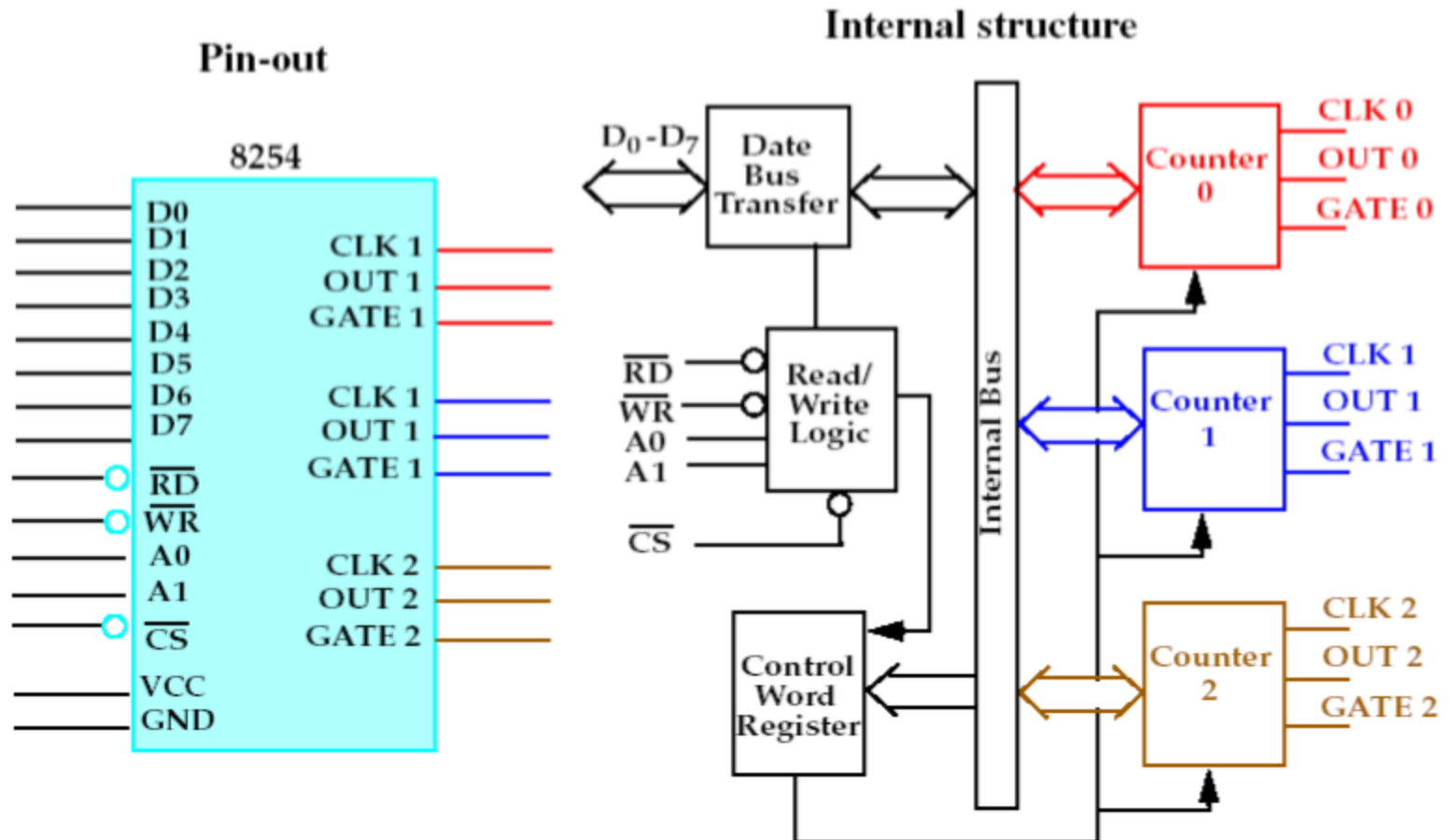
8253	8254
Its operating frequency is 0 - 2.6 MHz	Its operating frequency is 0 - 10 MHz
Read-Back command is not available	Read-Back command is available
Reads and writes of the same counter cannot be interleaved.	Reads and writes of the same counter can be interleaved.

Features of 8253 / 54

The most prominent features of 8253/54 are as follows

- It has three independent **16-bit down counters**.
- It can handle inputs from **DC to 10 MHz**.
- These three counters can be programmed for either **binary or BCD count**.
- It is **compatible** with almost all microprocessors.
- 8254 has a powerful command called **READ BACK** command, which allows the user to check the count value, the programmed mode, the current mode, and the current status of the counter.

Block Diagram of 8254



- Inside the 8253/54 timer, there are 3 counters.
- Each timer works independently and programmed separately.
- Each counter is assigned an individual port address.
- The control register common to all 3 counters and has its own port.

A ₁	A ₀	Result
0	0	Counter 0
0	1	Counter 1
1	0	Counter 2
1	1	Control Word Register

- Each of the three counter has 3 pins associated
 - CLK: input clock frequency
 - A square wave of 33% duty cycle
 - 8253: 0 ~ 2 MHz, 8254: 0 ~ 10 MHz
 - OUT: can be square wave, or one shot
 - GATE: Enable (high) or disable (low) the counter
- Data Pins: (D0 ~ D7)
 - Allow the CPU to access various registers inside the 8253/54 for both read and write operations. RD and WR are connected to IOR and IOW of control bus.

Example

- Interface 8253/54 with lower data byte of 8086 such that address of the counter 0, 1 ,2 and CW is 0008 h,000Ah, 000Ch and 000E h respectively.

Data given

- Counter 0: 0000 0000 0000 0000 1000
- Counter 1: 0000 0000 0000 0000 1010
- Counter 2: 0000 0000 0000 0000 1100
- Cntrlword: 0000 0000 0000 0000 1110

Lower byte of data of 8086.

Questions: A0 and A1 of 8253/54???

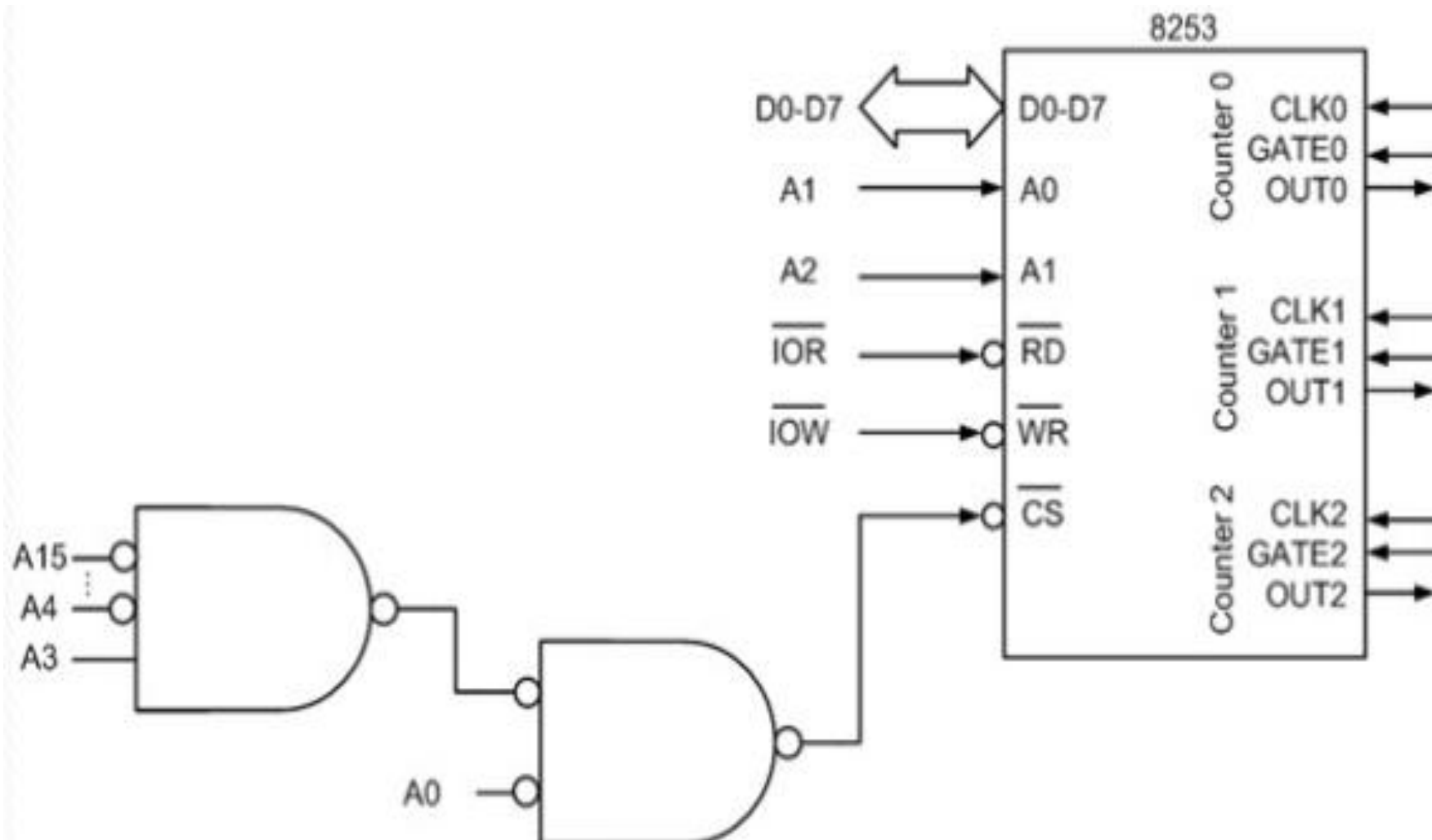
D0-D7 of 8253/54??

CS??

Gate 0, 1, 2 ??

Read line?? Writeline??

- Counter 0:0000 0000 0000 0000 1000
- Counter 1:0000 0000 0000 0000 1010
- Counter 2:0000 0000 0000 0000 1100
- Cntrlword:0000 0000 0000 0000 1110



Control word

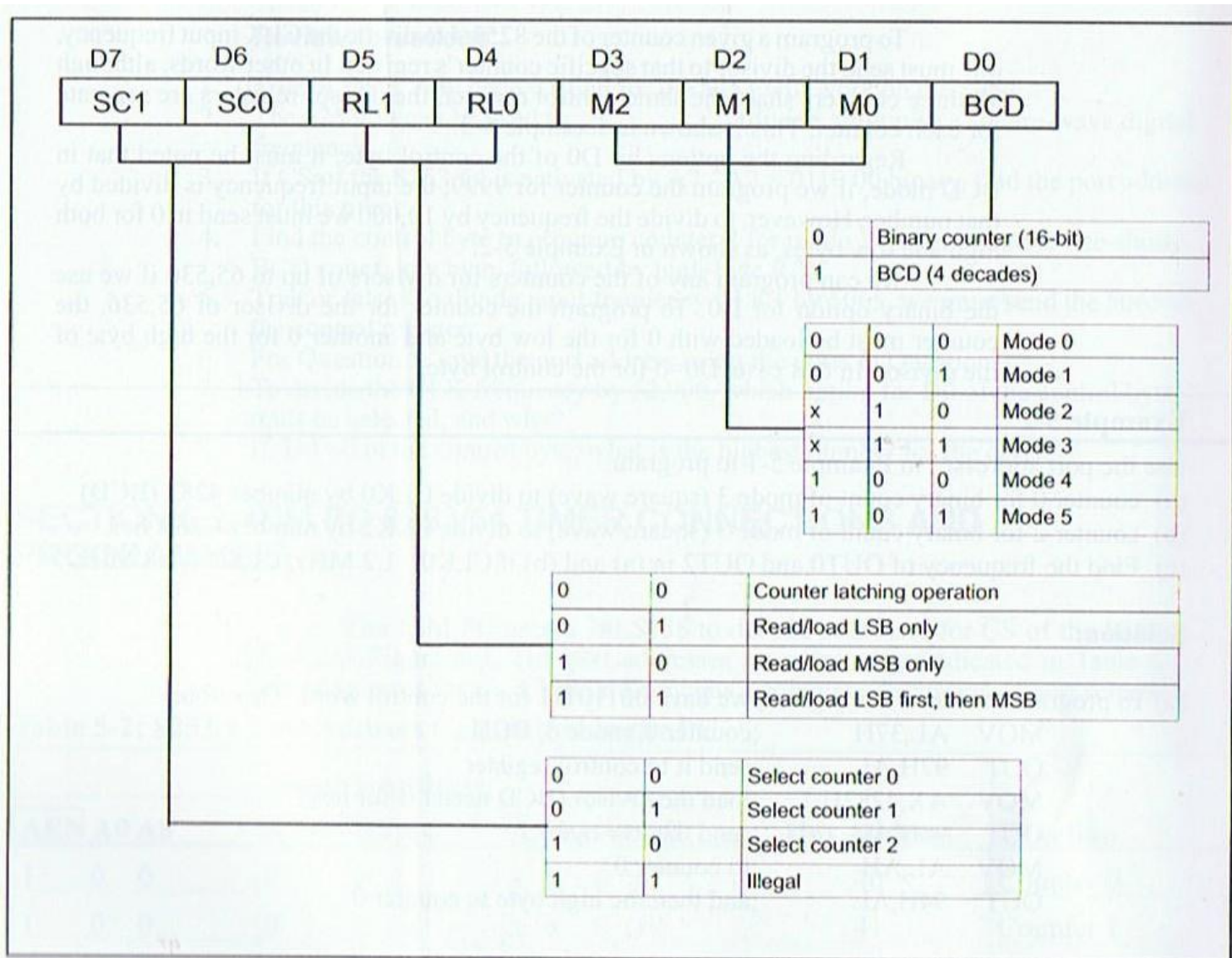


Figure 5-2. 8253/54 Control Word Format
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8253 / 8254 Timer

- Control byte must be first written into the control register. The 8253/54 must be initialized before use
- The programmer can not only write the value of the divisor into the 8253/54, but read the content of the counter at any given time as well
- All counters are down counters.

8253 / 8254 Timer

- To program a given counter to divide the CLK input frequency, one must send the divisor to that specific counter's register.
- Although all three counters share the same control register, the divisor registers are separate for each counter

8253 / 8254 Timer

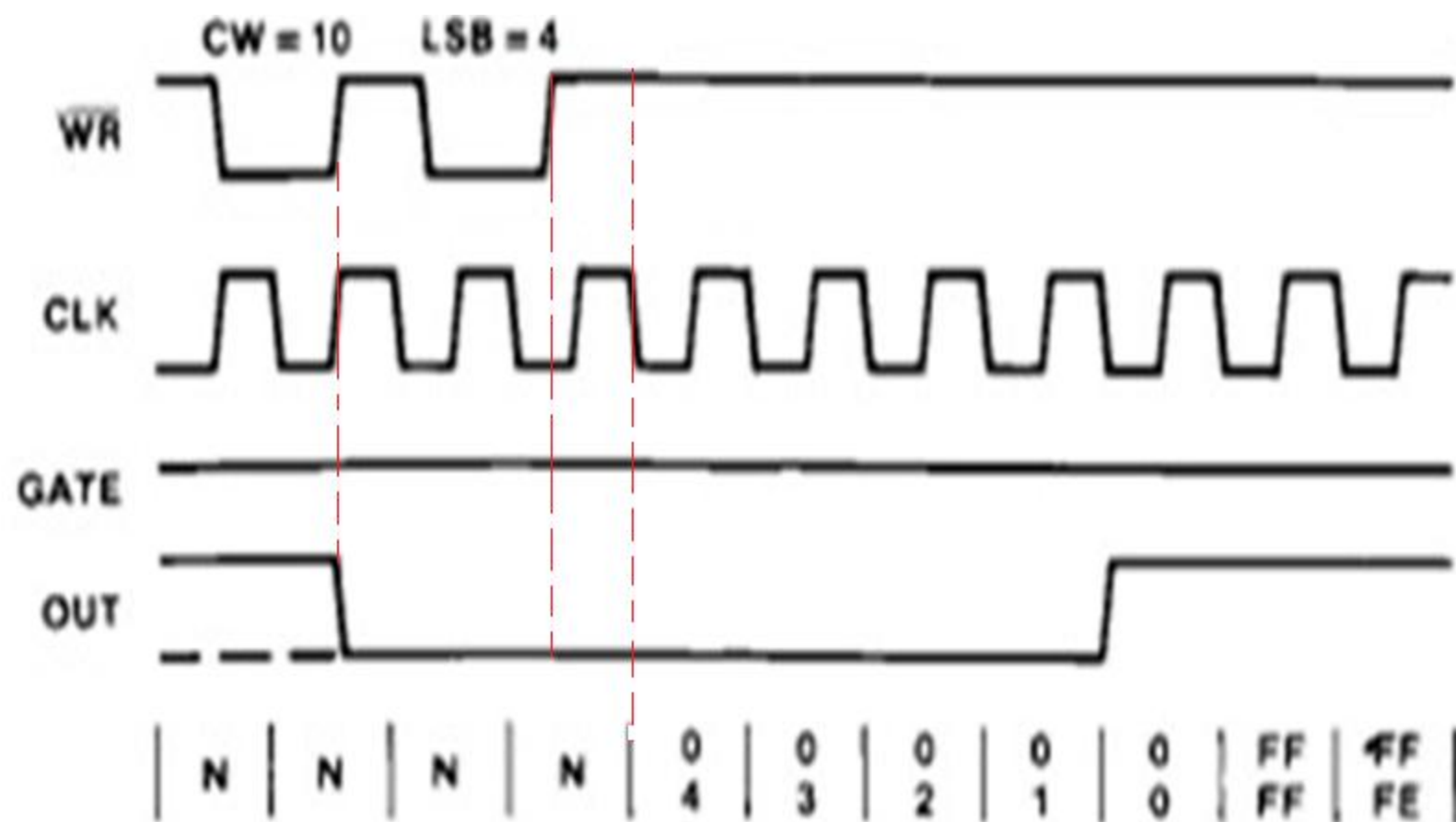
- Example: given the port addresses for 8253/54:
- Counter 0: 84H Counter 1: 85H
Counter 2: 86H Control Reg: 87H
- Task1: program counter 0 for binary counter for mode 3 to divide CLK0 by number 4282 (BCD)
MOV AL, 0011 0111B
OUT 87H, AL
MOV AX, 4282H
OUT 84H, AL (Low Byte)
MOV AL, AH
OUT 84H, AL (High Byte)
- $OUT0 = CLK0 / 4282$

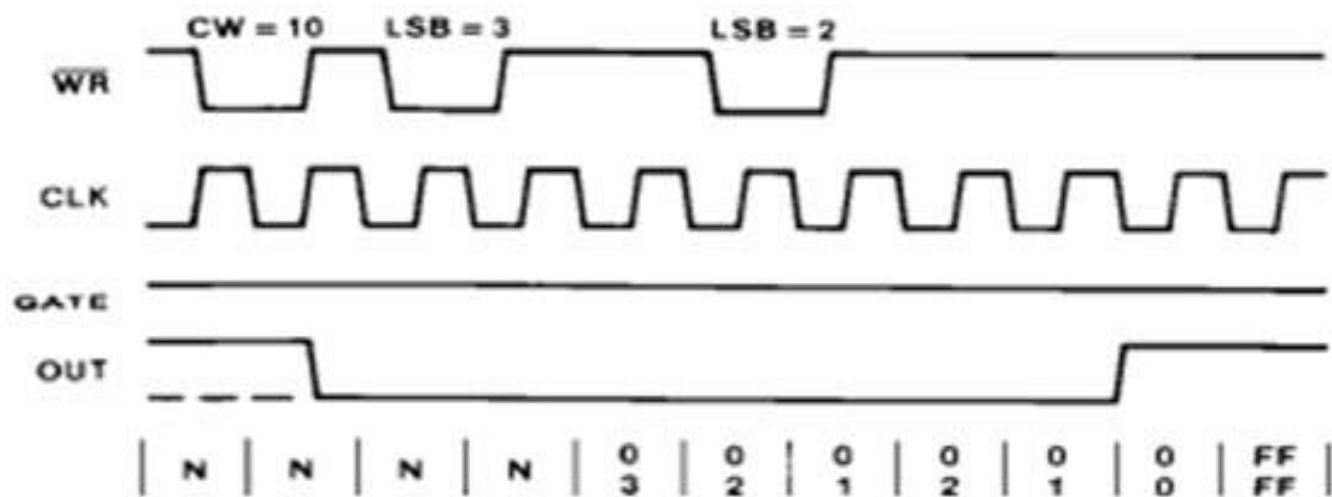
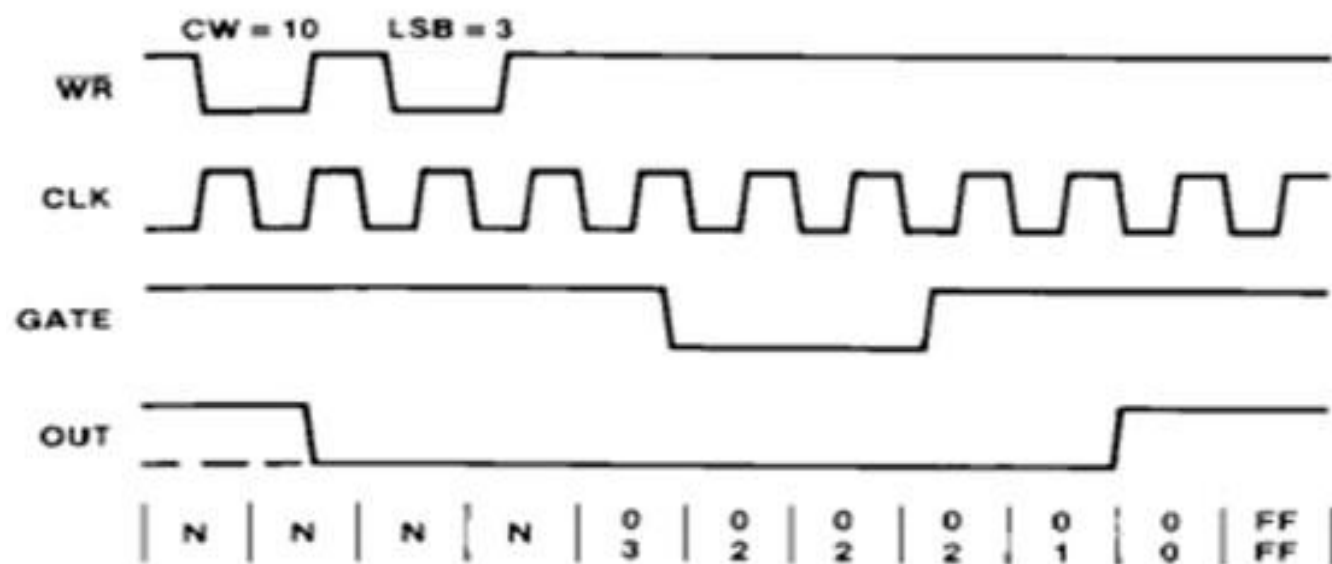
Shape of the 8253/54 Output

- Given CLK = 1.193 MHz, the clock period of input frequency is 838 ns
- If the number N loaded into the counter is even, both high and low pulse are the same length, which is $N/2 * 838 \text{ ns}$
- If the number N loaded into the counter is odd, the high pulse is $(N+1)/2 * 838 \text{ ns}$ and the low pulse is $(N-1)/2 * 838 \text{ ns}$
- ➔ If N is odd, the high portion of the output square wave is slightly wider than the low portion

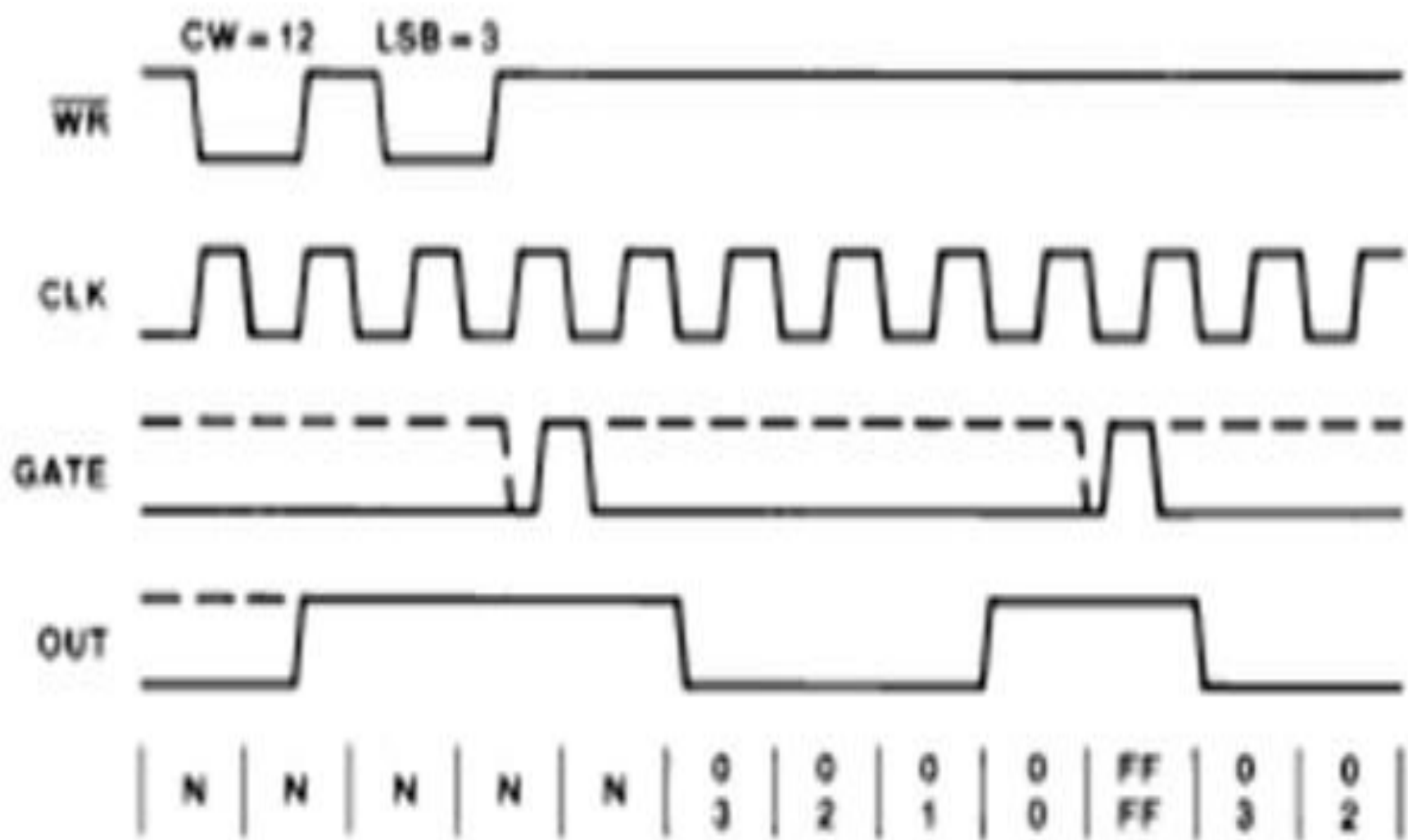
8253/54 Operation Modes

- Mode 0: Interrupt on terminal count
 - If GATE becomes low at the middle of the count, the count will stop and the output will be low. The count resumes when the GATE becomes high again → This in effect adds to the total time the output is low.

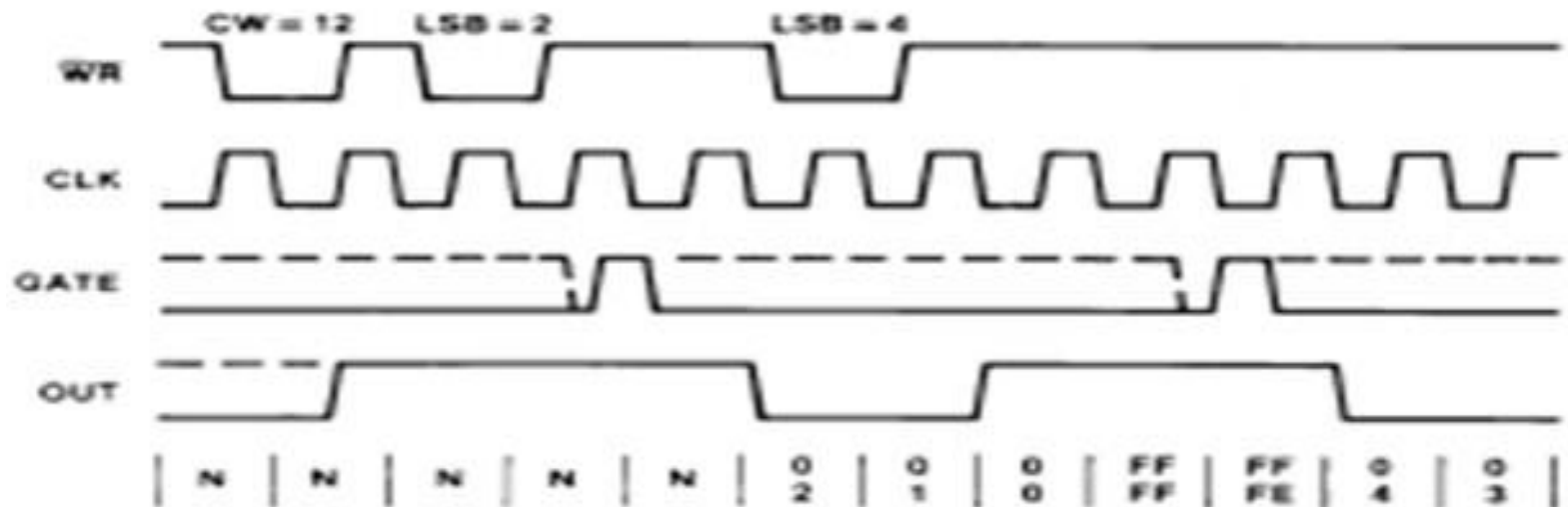
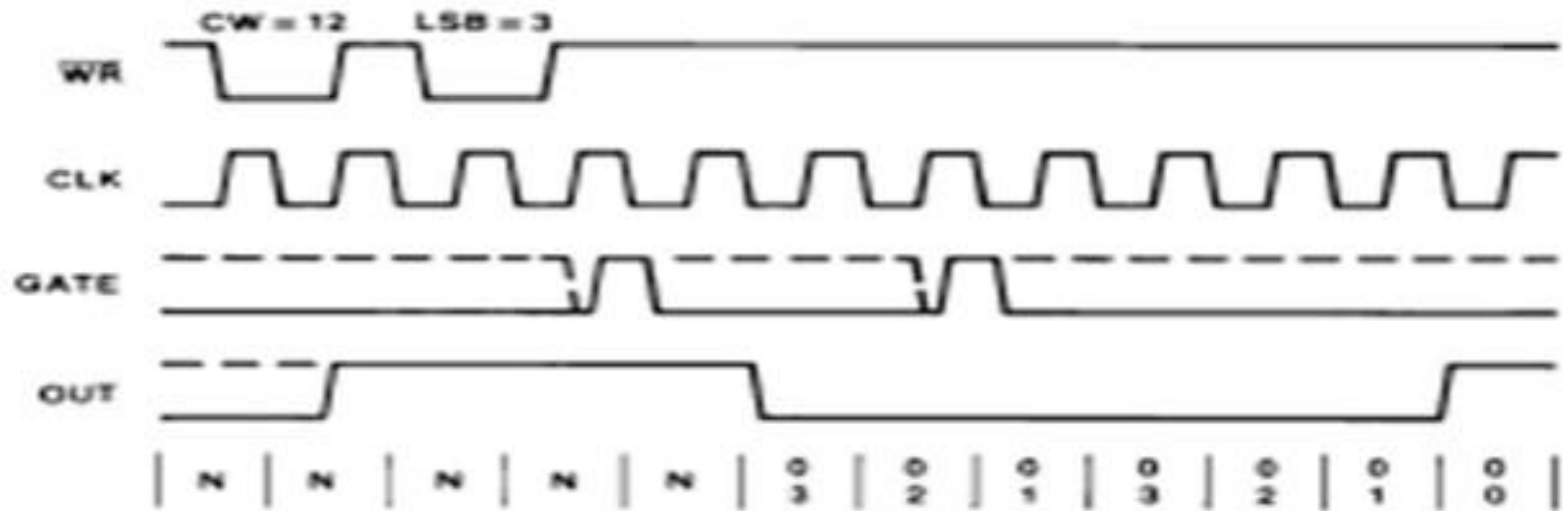




- **Mode 1: HW triggered / programmable one shot**
 - The triggering must be done through the GATE input by sending a 0-to-1 pulse to it.
 - Steps:
 - 1) Load the count register
 - 2) A 0-to-1 pulse must be sent to the GATE input to trigger the count
 - In Mode 1, after sending the 0-to-1 pulse to GATE, OUT becomes low and stays low for a duration of $N \cdot T$, then becomes high and stays high until the GATE is triggered again
 - If during the activation, a retriggered happened, then restart the down counting

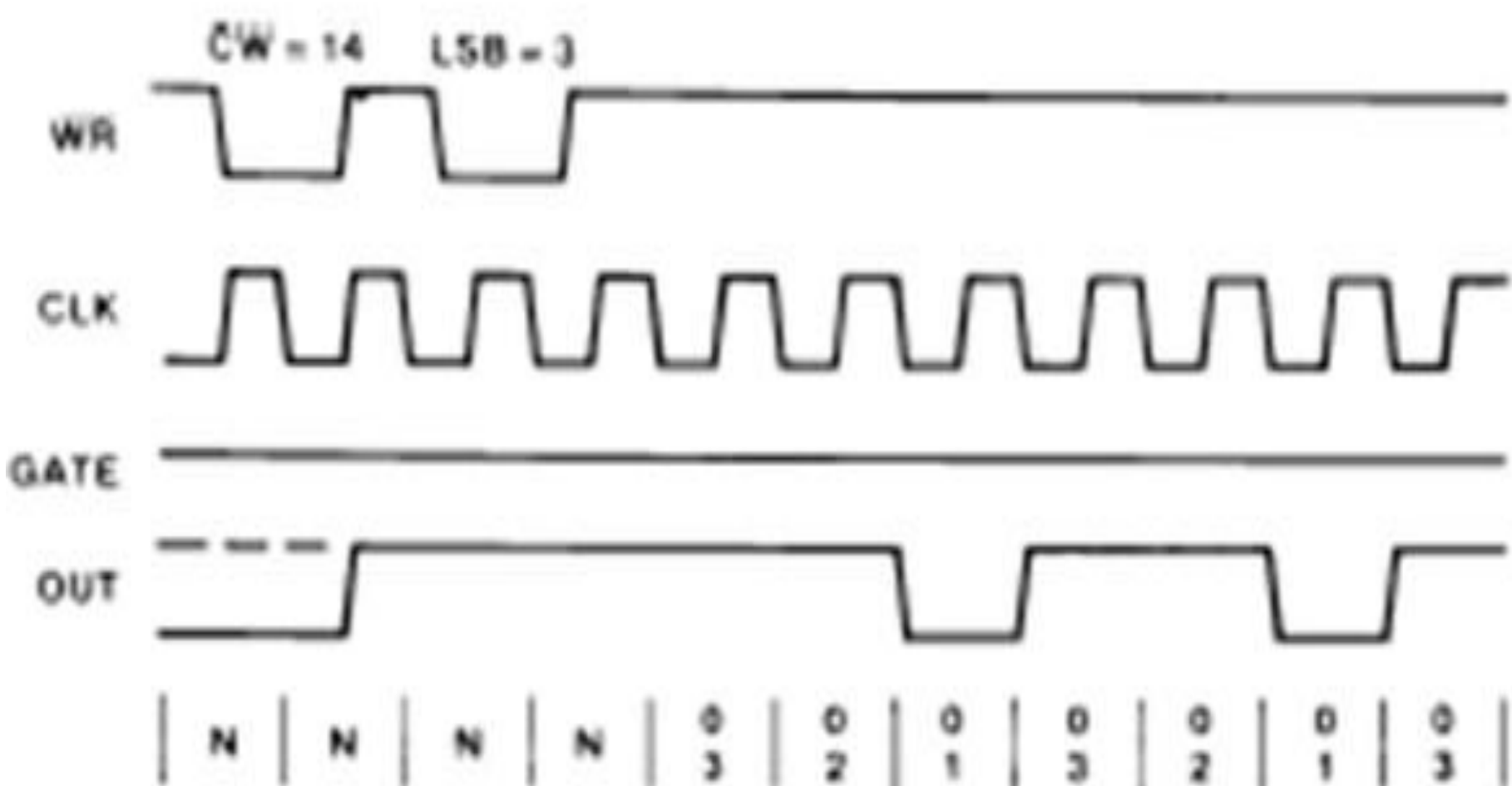


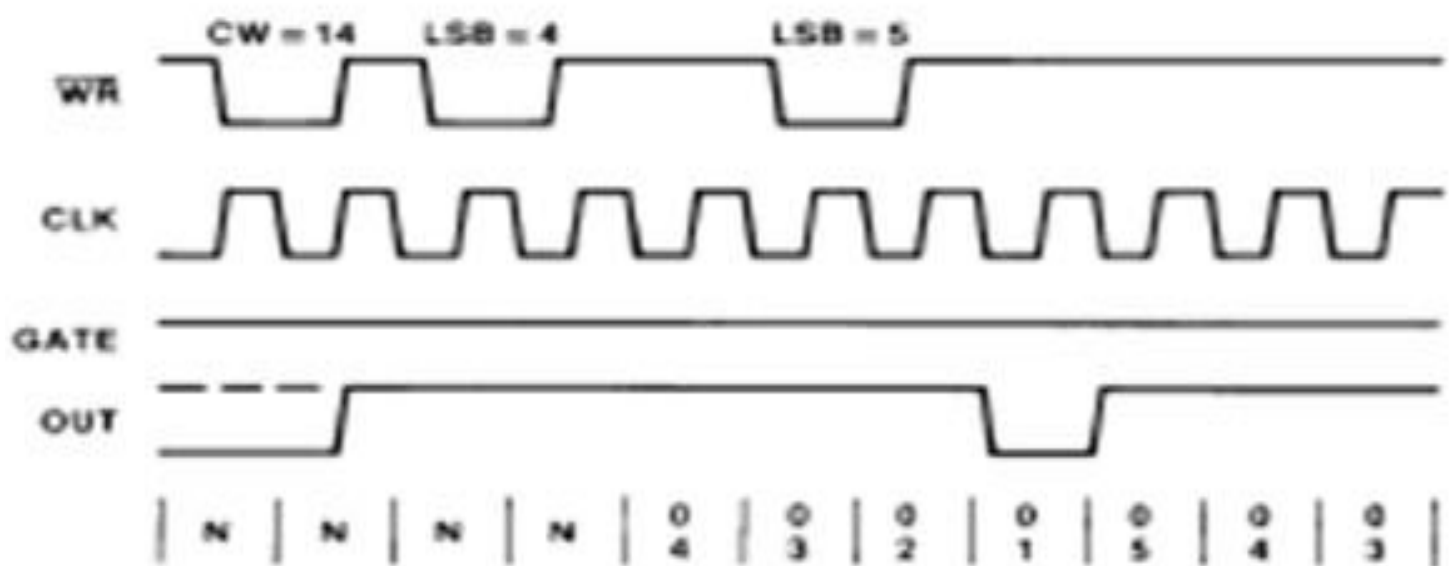
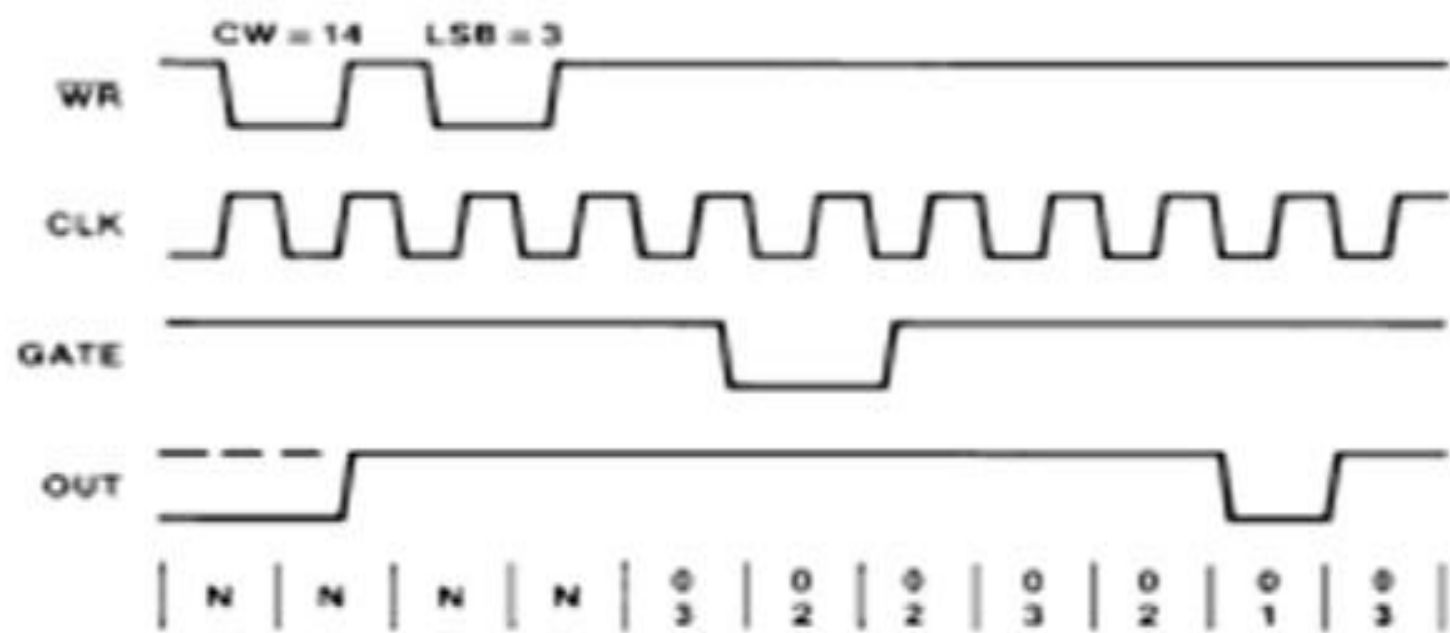
Mode 1



8253/54 Operation Modes

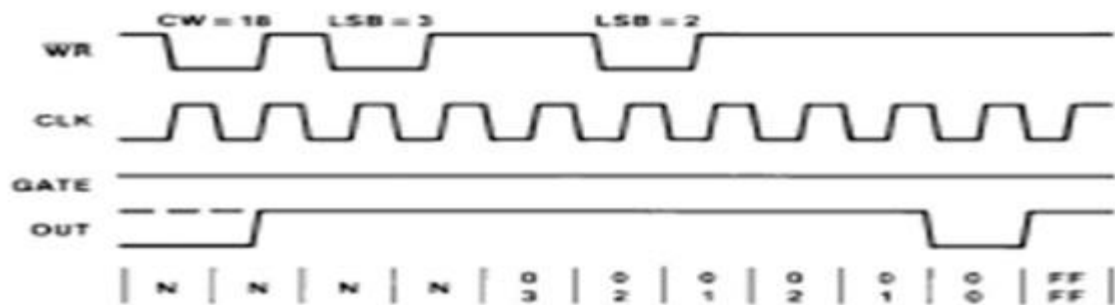
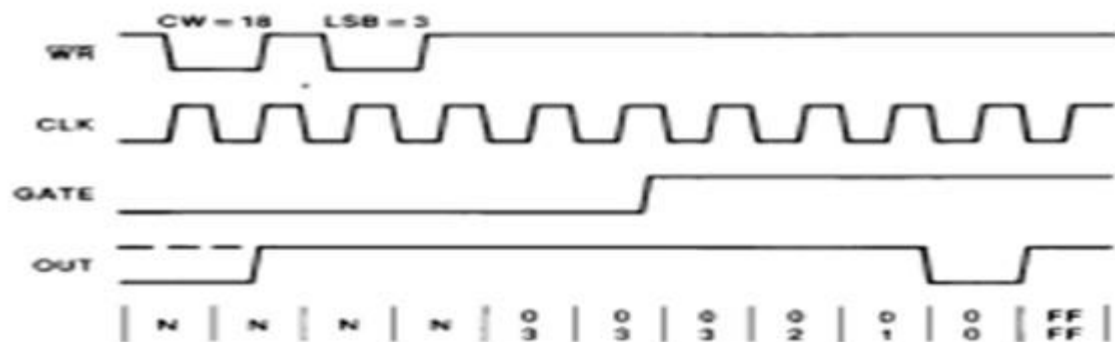
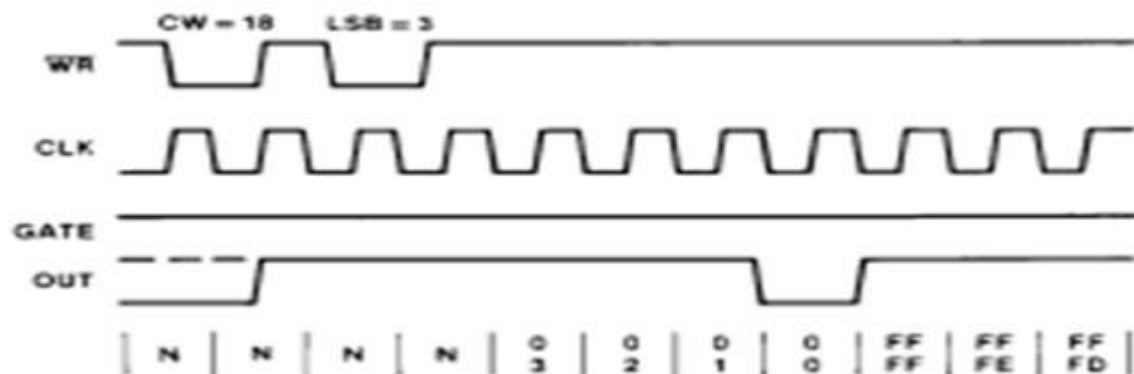
- Mode 2: Rate Generator (Divide-by-N counter)
 - In Mode2, if GATE=1, OUT will be high for $(N-1)*T$, goes low only for one clock pulse, then counter is reloaded automatically, and the process continues indefinitely. ➔ Whole period: $N * T$





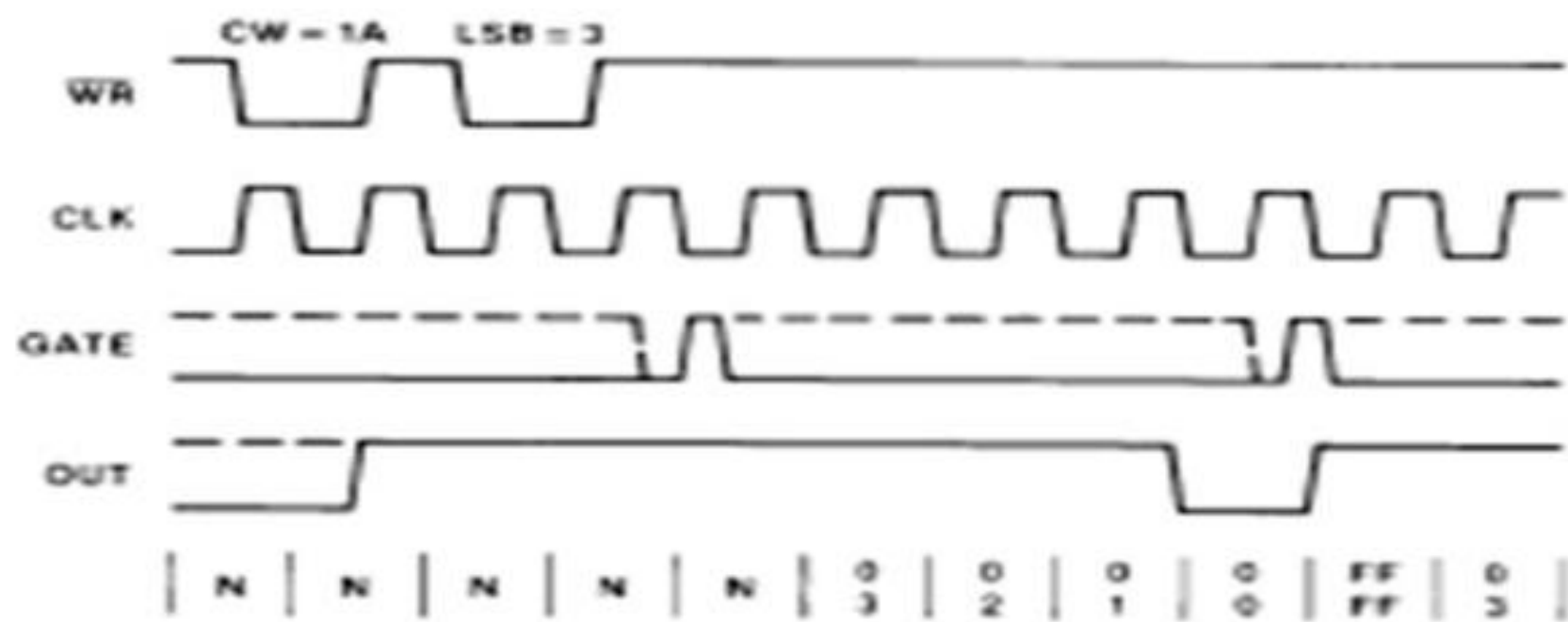
Mode 4: Software triggered strobe

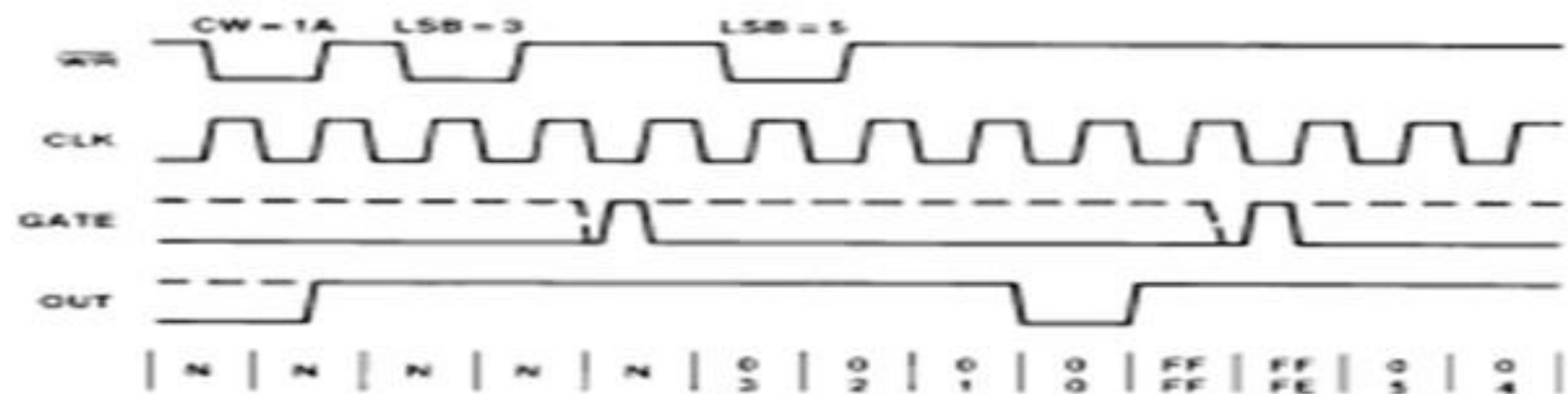
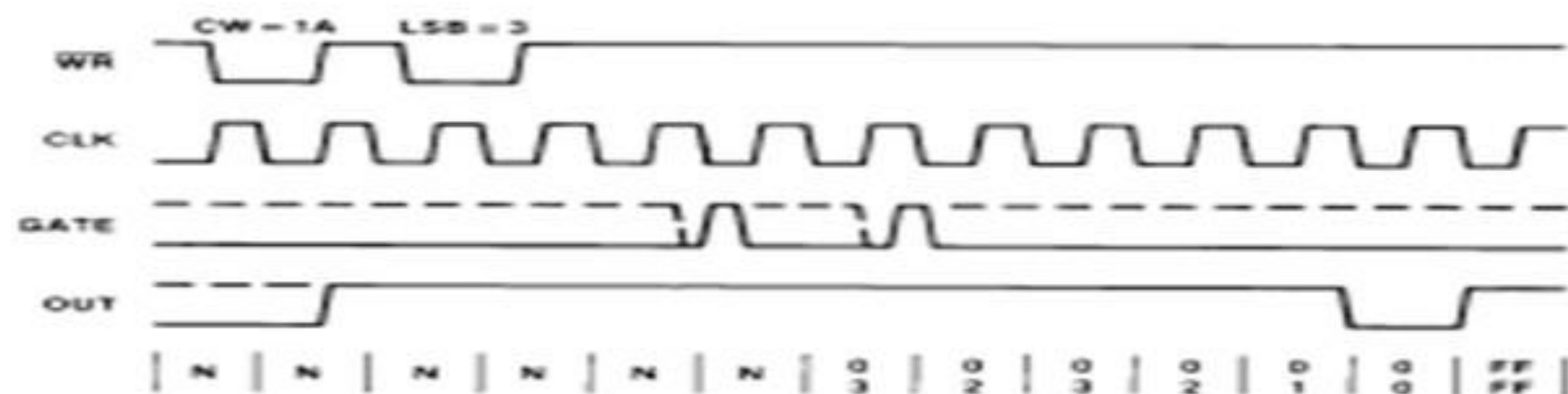
- Similar to Mode2, except that the counter is not reloaded automatically
- In Mode4, if GATE=1, the output will go high when loading the count, it will stay high for duration $N \cdot T$. After the count reaches zero, it becomes low for one clock pulse, then goes high again and stays high until a new command word or new count is loaded
- To repeat the strobe, the count must be reloaded



8253/54 Operation Modes

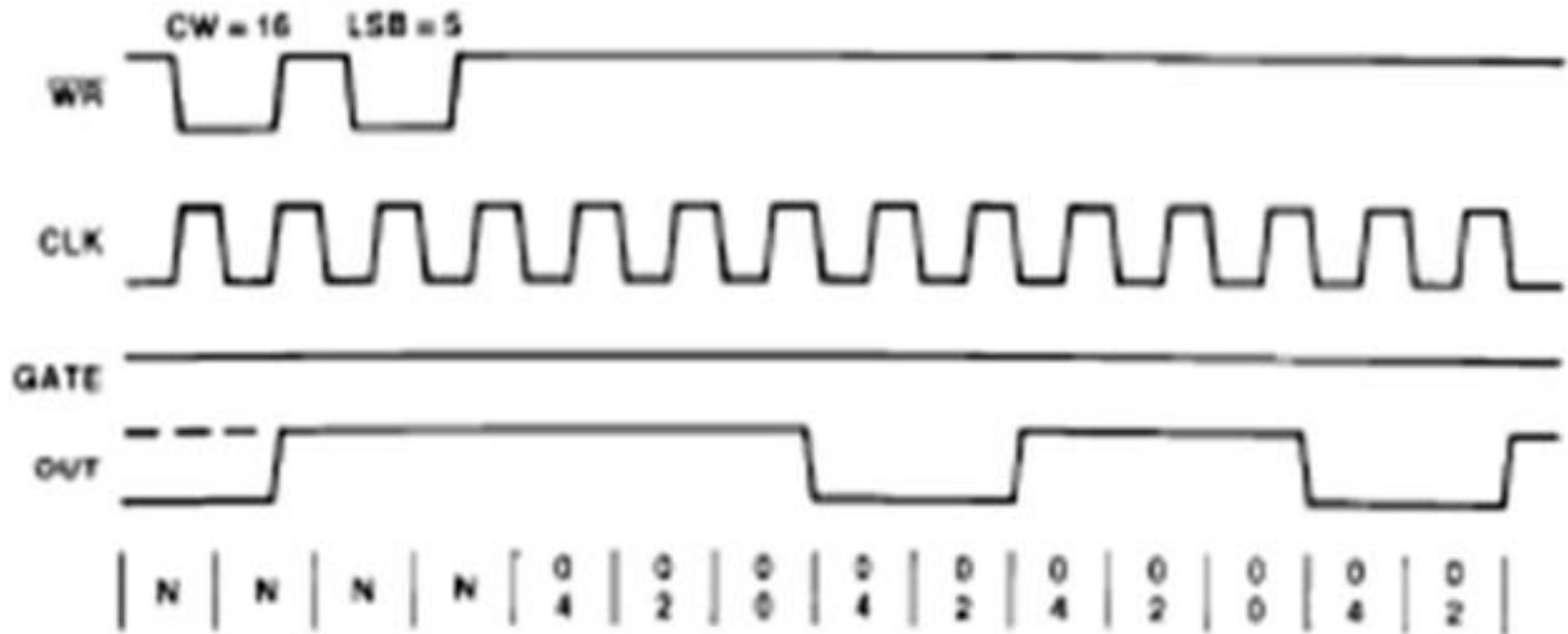
- Mode 5: Hardware triggered strobe
 - Similar to Mode4, except that the triggering must be done with the GATE input
 - The count starts only when a 0-to-1 pulse is sent to the GATE input
 - If GATE retriggered during the counting, it will restart the down counting

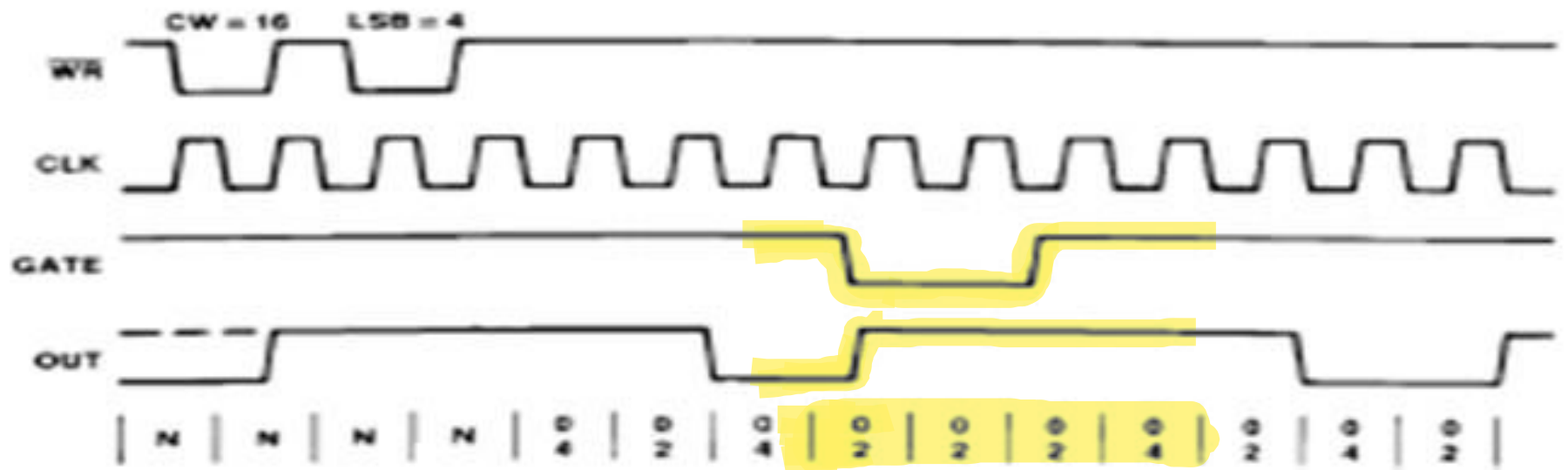
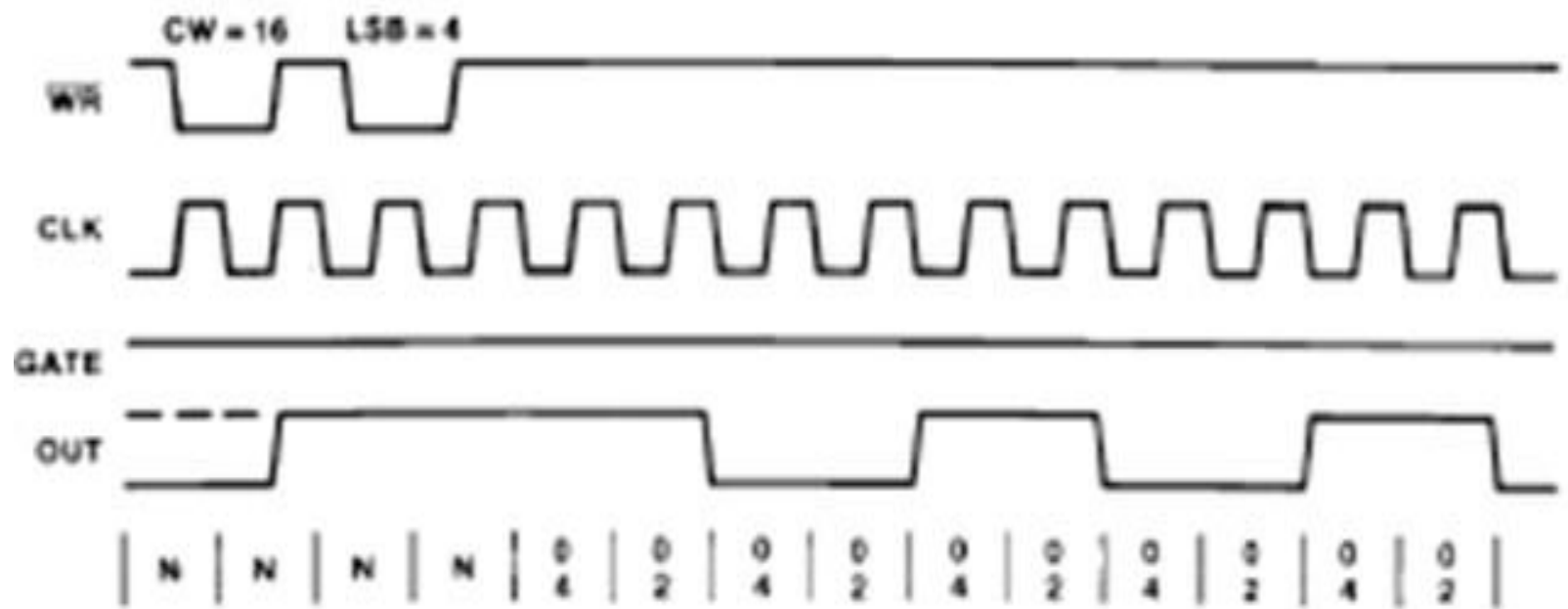




8253/54 Operation Modes

- **Mode 3: Square wave rate generator**
 - Most commonly used





8254 Read Operations

- There are three possible methods for reading the counters:
 - A simple read operation
 - The Counter Latch Command, and
 - The Read-Back Command.

Simple Read Operation

- This operation read the counter after stopping.
- To read the Counter, which is selected with the A1, A0 inputs, the CLK input of the selected Counter must be inhibited by using either the GATE input or external logic. Otherwise, the count may be in the process of changing when it is read, giving an undefined result.
- Two I/O read operation are performed by the MPU
 1. The first I/O operation reads the low order byte.
 2. The second I/O operation reads high order byte.

Counter Latch Command

- This allows reading the contents of the Counters “on the fly” without affecting counting in progress.
- The selected Counter's output latch (OL) latches the count at the time the *Counter Latch Command* is received.
- This count is held in the latch until it is read by the CPU (or until the Counter is reprogrammed). The count is then unlatched automatically and the OL returns to “following” the counting element (CE).

Counter Latch Command (Continued)

Counter Latch Commands do not affect the programmed Mode of the Counter in any way.

Example:

; Latching counter0

MOV DX, C_REG

MOV AL, 00000000B ; count latched for counter 0.

OUT DX, AL

; Reading counter0

MOV DX, CNTR0

IN AL, DX

$A_1, A_0 = 11$; CS = 0; RD = 1; WR = 0

D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀
SC1	SC0	0	0	X	X	X	X

SC1, SC0—specify counter to be latched

SC1	SC0	Counter
0	0	0
0	1	1
1	0	2
1	1	Read-Back Command

D₅, D₄—00 designates Counter Latch Command

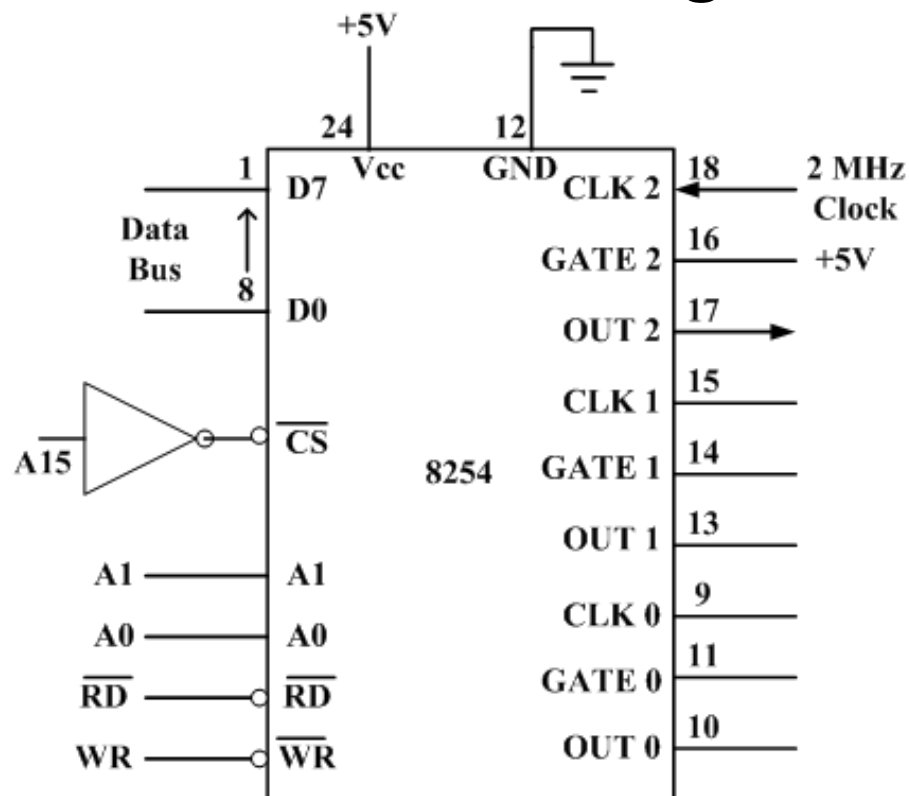
X—don't care

NOTE:

Don't care bits (X) should be 0 to insure compatibility with future Intel products.

Problem

- Identify the port address of the control register and counter 2 in figure.



Problem

- Write instructions to generate a 1 KHz square wave from Counter1. Assume the gate of counter1 is tied to +5V through a 10K resistor. (Clock=2MHz)

Soln: To generate a square wave from counter1, it should be initialize in Mode 3.

Needed count = 2000 (i.e $2\text{MHz} / 1\text{KHz}$)= 07D0H

Control Word=76H

0	1	1	1	0	1	1	0
---	---	---	---	---	---	---	---

```
CNT1 EQU 8001H
CNTR EQU 8003H
MOV  AL, 76H
OUT  CNTR, AL
MOV  AL, D0H
OUT  CNT1, AL
MOV  AL, 07H
OUT  CNT1, AL
```